Sirou Zhu

Generative AI | Post-Training and RL | Agent Systems

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Research scientist and engineer specializing in post-training and reinforcement learning for multimodal foundation models, tool-using agents, and physical AI. Builds the full stack from synthetic data and reward modeling to large-scale training, evaluation, and efficient deployment.

TECHNICAL FOCUS

Post-training: SFT, reinforcement learning, preference optimization, reward modeling, distillation, synthetic data, and agent evaluation

Models and systems: LLMs, VLMs, VLAs, tool-use agents, PyTorch, TensorFlow, verl, LangGraph, distributed training and inference

EXPERIENCE

Luma AI | Research Scientist / Engineer, Foundation Models

Apr. 2026 - Present

Agent as a Policy for Physical AI AGENTS | VLM POST-TRAINING | RL

- Developed an agent-as-policy framework for robotic manipulation that combines tree-search data generation with efficient VLM post-training through SFT and RL, achieving state-of-the-art simulation performance and enabling complex real-world manipulation.
- Optimized reinforcement-learning algorithms and infrastructure for efficient policy post-training.
- Trained a vision-language-action model as a specialized tool for hierarchical VLM-VLA collaboration.

Multimodal Design Agent for Infographic Generation GENERATION | RL | REWARD MODELING

- Post-trained a VLM with RL to generate polished HTML infographics and developed a reward model aligned with human aesthetic preferences.

LinkedIn | Staff Research Engineer, Foundation AI

Aug. 2025 - Apr. 2026

LinkedIn | Senior Research Engineer, Foundation AI

Aug. 2024 - Aug. 2025

Cross-Episode Meta-RL and Tool-Use Agents AGENTIC RL | SYNTHETIC DATA | EVALUATION

- Improved LLM in-context online learning with multi-task, multi-episode reinforcement learning.
- Built graph- and MCTS-based synthetic data generation for ambiguous multi-hop queries, plus finite-state-machine generation that drives tool-calling state transitions for clear-intent queries.
- Created a rubric-based agent-as-judge evaluation framework with more than 80% human agreement, enabling reliable automatic quality measurement.
- Improved hallucination and logical reasoning with agentic RL; built a training loop in verl to reduce tokenization drift and improve stability.

Preference Optimization and Model Efficiency DPO | ORPO | SIMPO | SPPO | DISTILLATION

- Built a preference-optimization pipeline spanning DPO, ORPO, SimPO, and SPPO; developed AlphaPO, an alpha-divergence extension of SimPO that delivered 7%-10% alignment gains on Mistral and Llama.
- Designed a reasoning-model distillation recipe that retained downstream performance while truncating 50% of rationale data, reducing training time and memory usage; contributed to the open-source FMChisel library.

360Brew Production LLM TRAINING SYSTEMS | INFERENCE EFFICIENCY

- Customized attention masks and positional encoding for multi-label scoring, improving inference efficiency by 3x without hurting performance; optimized data-mixing strategies for final relevance.

SELECTED GENERATIVE AI RESEARCH

Scaling In-Context Online Learning Capability of LLMs via Cross-Episode Meta-RL - Preprint, 2026

Scaling Down, Serving Fast: Compressing and Deploying Efficient LLMs for Recommendation Systems - EMNLP Industry, 2025

Efficient AI in Practice: Training and Deployment of Efficient LLMs for Industry Applications - Preprint, 2025

AlphaPO: Reward Shape Matters for LLM Alignment - ICML, 2025

EARLIER EXPERIENCE

LinkedIn | Senior Machine Learning Engineer, Ads Ranking Jan. 2023 - Aug. 2024

- Modeled long-term reward-to-go and added a decision-transformer module for member-activity modeling in the ads click-through-rate model.
- Initiated and led personalized multimodal GNN development across onsite ads, offsite ads, and personalized ad creation; established graph generation, training, and embedding-inference pipelines used across teams.
- Introduced incremental learning for transductive ID-based GNNs and led min-max quantization that reduced embedding-table size by more than 70%.
- Applied low-rank DCNv2 and transformer-based member-history modeling for large-scale personalized ranking.
- Incorporated ads revenue into a notification reinforcement-learning system, increasing revenue without increasing notification volume.
- Built a cross-domain modeling pipeline that integrated feed signals into ads ranking and improved relevance and click-through rate.

LinkedIn | Machine Learning Engineer, Ads Ranking Aug. 2021 - Jan. 2023

- Built a TensorFlow deep-learning framework adopted by multiple ads teams, broadening production model support and improving click-through-rate and revenue metrics.
- Developed automatic retraining infrastructure and accelerated training by 8x through model compression and indexing optimization.
- Developed multi-behavior and multi-task learning models around member-attention signals, launching four models with substantial click-through-rate gains.

LinkedIn | Machine Learning Engineer Intern May 2020 - Aug. 2020

- Developed a distributed PyTorch L-BFGS optimizer with stochastic progressive batching and line search, converging 20% faster on w8a and 200% faster on a9a than regular L-BFGS.
- Removed the legacy data-size limitation in regular L-BFGS, making the method applicable to large-scale datasets.

SELECTED SYSTEMS PUBLICATIONS

Efficient Algorithms for Leveraging LLMs for Generative and Predictive Recommender Systems - WWW, 2025
LiGNN: Graph Neural Networks at LinkedIn - KDD Best Paper, 2024
LiRank: Industrial Large-Scale Ranking Models at LinkedIn - KDD, 2024
Logit Attenuating Weight Normalization - Research publication

EDUCATION

Carnegie Mellon University | M.S. Computational Data Science, GPA 3.91/4.00 2019 - 2021
Zhejiang University | B.E. Software Engineering, GPA 3.89/4.00 2015 - 2019